Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2026

Course: B.Tech | *Subject: Digital Electronics* | *Time: 03 Hours*
Code: 103401 | | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) Simplify $F(A, B, C, D) = \Sigma m(0,2,8,10)$
(i) $B'D'$ (iii) BD
(ii) $A'C'$ (iv) $B'D$

(b) How many 4:1 multiplexers are needed to build a 16:1 multiplexer?
(i) 3 (iii) 4
(ii) 5 (iv) 8

(c) The minimum number of flip-flops required for a mod-12 counter is
(i) 3 (iii) 4
(ii) 5 (iv) 12

(d) Which pair of gates is called universal gates?
(i) AND, OR (iii) NOT, AND
(ii) NAND, NOR (iv) XOR, XNOR

(e) A T flip-flop can be obtained from a JK flip-flop by connecting
(i) $J = K = T$ (iii) $J = 0, K = T$
(ii) $J = T, K = T'$ (iv) $J = T, K = 0$

(f) The main property of Gray code is that
(i) it is a weighted code (iii) it is self-complementing
(ii) successive numbers differ in only one bit (iv) it uses 4 bits only for decimal digits

(g) The fastest type of ADC is
(i) dual slope (iii) successive approximation
(ii) counting (iv) parallel (flash)

(h) An 8-bit successive approximation ADC uses a 1 MHz clock. The conversion time is:
(i) 1 μs (iii) 8 μs
(ii) 255 μs (iv) 256 μs

(i) CMOS has very low static power dissipation because
(i) it operates at a low voltage only (iii) in steady state, only one of the complementary transistors conducts
(ii) it uses no transistors (iv) it has high fan-out

(j) EPROM is erased using
(i) electrical pulses (iii) magnetic field
(ii) ultraviolet light (iv) heat

Q.2 (a) Minimise $F(A,B,C,D) = \Sigma m(0,1,2,5,6,7,8,9,10,14)$ using a K-map. Give the minimal SOP and draw the NAND-only implementation. [7]

(b) Design a synchronous 3-bit counter with sequence $0 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 5 \rightarrow 0$ using T flip-flops. Handle the unused states so that the counter is self-starting. [7]

Q.3 (a) Use the Quine-Mc Cluskey method to minimise $F(A,B,C,D) = \Sigma m(0,1,2,5,6,7,8,9,10,14)$. [7]

(b) Classify semiconductor memories. Define volatile, non-volatile, random access, sequential access, access time and cycle time. [7]

Q.4 (a) A 7-bit Hamming code is received as 1011011 (even parity, positions numbered from the left). Locate and correct the error and write the original 4-bit data word. [7]

(b) Explain the dual slope ADC and show that its output is independent of the clock frequency and the integrator components. [7]

Q.5 (a) Explain why NAND and NOR are called universal gates. Show how each can realise AND, OR and NOT operations [7]

(b) Implement the following using a PAL: $F1 = AB + A'C, F2 = A'B' + BC$. Show the fuse map and diagram. [7]

Q.6 (a) Explain the content addressable memory. Analyse how it differs from RAM in search operations and where it is used. [7]

(b) Perform $45 - 29$ and $29 - 45$ in 8-bit binary using (i) 1's complement and (ii) 2's complement arithmetic. Comment on how overflow is detected. [7]

Q.7 (a) Implement $F(A,B,C) = \Sigma m(1,2,4,7)$ using (i) an 8:1 MUX and (ii) a 4:1 MUX. Draw both circuits. [7]

(b) Define resolution, accuracy, linearity, settling time, offset error and monotonicity of a D/A converter. [7]

Q.8 (a) Describe the working of a standard TTL NAND gate with a totem-pole output. Explain how the Schottky diode and transistor improve its speed. [7]

(b) Analyse the race-around condition in a level-triggered J-K flip-flop and explain how master-slave and edge-triggering solve it. [7]

Q.9 Write short note on ***any two*** of the following: [7 x 2 = 14]

(a) Implement a full adder using two half adders and an OR gate

(b) shift registers

(c) Convert JK flip-flop into a D flip-flop using the excitation table

(d) Justify the need for a sample-and-hold circuit before an ADC